

INTRODUCTION TO PHYSIOLOGY

- Physiology tells us **how the bodies of living organisms work.**
- Both **structure and function** must be studied at all levels from the **cellular to organism.**
- All aspects of human physiology evolved in the **thousands of inherited units of DNA called genes.**

Cont..

- There is immense **genetic diversity**, as a result of small spontaneous change in **individual genes, called mutation**
- The natural selection concept of **Charles Darwin** emphasizes the **predominance of the genes** in the population that **favors survival of the fittest and reproduction** in a particular environment.

Body Composition

- At an average, **60% of the body weight** of young adult male is **water**.
- The remaining is composed of **minerals, fat & proteins, carbohydrates and nucleic acids**.
- The **lipids** are important forms of **storage fuel** in addition to providing **insulation** of the body.

Cont...

- Proteins serve as the **structural basis for all enzymes, contractile muscle proteins, Connective tissue, collagen & elastin**
- Proteins are precursor for carbohydrate in the process of **gluconeogenesis**.
- Ingested glucose is converted to **glycogen** and stored in the **liver, muscle and adipose tissue**.

HOMEOSTASIS

- **a dynamic equilibrium** where body conditions are **maintained within narrow limits**
- 1. necessary for each cell to survive
- 2. each cell contributes
- 3. all cells are in contact with the aqueous (watery) internal environment, connects all cells,

Cont...

- exchanges made is a delicately **balanced state**
- Many of these regulatory mechanisms operate on the **negative feedback**.
- The **structure and chemical reactions** of living organisms are sensitive to the chemical and physical conditions **within and around cells**.
- The ICF has a high concentration of K^+ and low concentration of Na^+ , Cl^- , Mg^{++} , and Ca^+ .

Cont...

- Cells need homeostasis for **their own survival** and for performing **specialized function essential to survival of the whole body.**
- Cells need a constant supply of nutrient and oxygen and ongoing elimination of acid-forming carbon dioxide, to generate energy needed to **power life sustaining cellular activities.**
- **Food + Oxygen = Carbondioxide + water + Energy**

Major Organ Systems involved in homeostasis

.Controlled mainly by **nervous and endocrine s.**

1. Sensor

- a. monitors variable (factor being regulated)
- b. responds to changes (stimuli) by sending input

2. Integrator

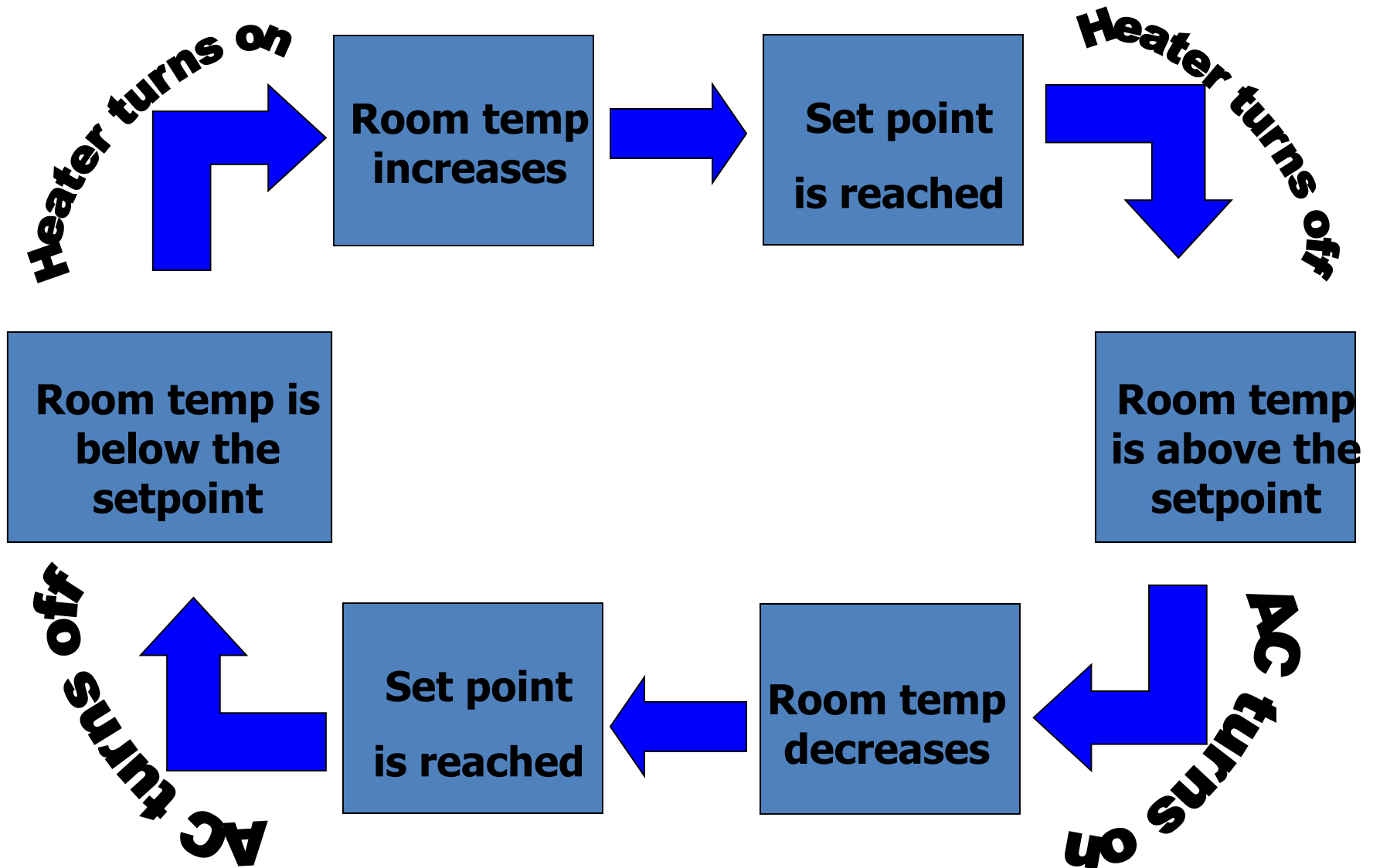
- a. determines set point
- b. compares set point to input
- c. sends response to...

3. Effector

- . responds to changes

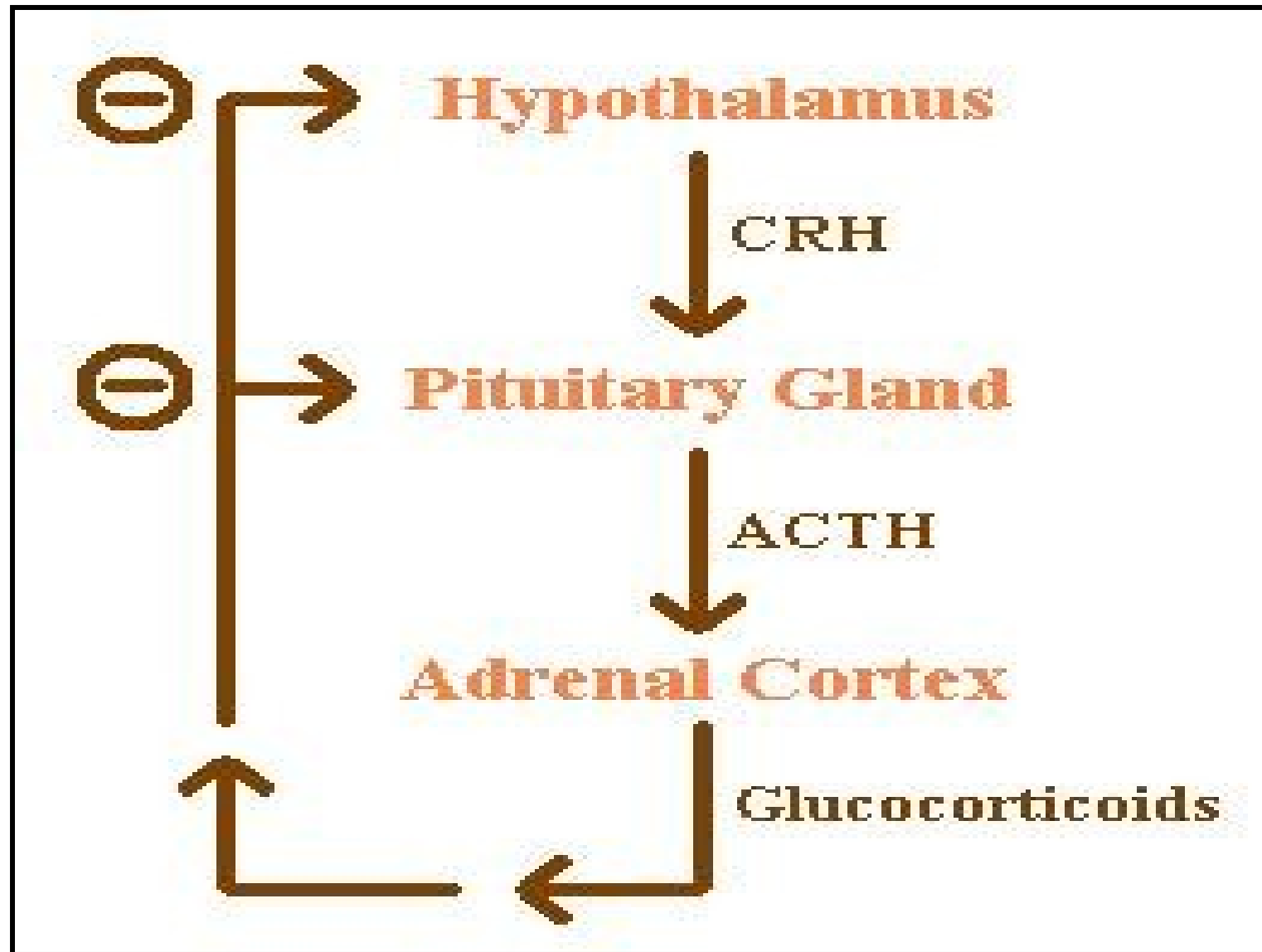
Negative Feedback

- Negative feedback occurs when the **output of a system acts to oppose changes to the input of the system.**
- A thermostat is an example of a negative feedback system.



A thermostat is a device for regulating the temperature of a system so that the system's temperature is maintained near a *desired setpoint temperature*.

Negative Feedback (Stress hormone)



Cont...

- The **brain** continuously sends signals to the endocrine glands to **secrete and release hormones** and the **glands, in turn, send feedback to the nervous system.**
- The **hypothalamus** in the brain is the **master switch** that sends signals to the **pituitary gland** which can release up to **eight hormones** into the bloodstream.

Cont...

- The hypothalamus then detects the rising hormone levels and decreases the release of hormones from the pituitary which results in **a decrease in hormone release from the target organ.**
- The process of maintaining normal body function through negative feedback mechanisms is called **homeostasis.**

Glucose and Insulin

- **Glucose** intake occurs during digestion of food that is needed for energy expenditure to perform routine physical activities.
- **The pancreas** is the key organ that regulates the glucose levels in body by secreting two hormones, **insulin and glucagon**.
- The liver also helps to store the excess glucose in form of **glycogen to be utilized later**.

Glucose and Insulin Negative Feedback Loop

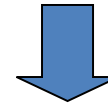
- Two primary Hormones

Insulin



Lowers Blood Glucose Levels

Glucagon



Raises Blood Glucose Levels

The opposite actions of these two hormones, insulin and glucagon, helps to maintain **normal blood sugar levels** in the body hence maintain homeostasis of the body.

ADH regulated water conservation in kidneys

**Less water in the
blood**



**Stimulates osmoreceptors in
hypothalamus to send signals
to the pituitary gland**



**Pituitary glands secretes
high levels of ADH**



**ADH makes the tubules more permeable
and more water is reabsorbed back into
the bloodstream**

**Excess water in the
blood**



**Stimulates osmoreceptors in
hypothalamus to send signals
to the pituitary gland**



**Pituitary glands secretes
low levels of ADH**



**Less ADH makes the tubules
less permeable and less
water is reabsorbed**

Cell to cell adhesions

- **carbohydrates on membrane surface help arrange cells into groups**

1. Cell adhesion molecules (CAMs)

2. extracellular matrix (connective tissues)

- Cells may not be joined directly to other cells, but embedded in matrix of **carbohydrates and protein fibers**

- (1) collagen - resists tension (e.g., skin)

- (2) elastin - stretch and recoil (e.g., skin, lungs)

- (3) fibronectin - holds cells in position (all over body)

3. specialized junctions

- **a. desmosomes** - cells join at particular spots, found all over body, particularly where stretch occurs (e.g., skin, muscle)
- **b. tight junctions** - impermeable barrier, common in epithelial sheets where they prevent leakage
- **c. gap junctions** - cells linked by protein tunnels, allows small molecules to pass between cells, important in some cells that transmit electrical activity (e.g., cardiac muscle)

Cellular Transport

- Cellular transport is the process of molecules or material **moving in and out of the cell**
- two types
- Passive Transport
- Active Transport

Passive Transport

- is the movement of molecules in or out of the cell **without the use of energy**
- Can only occur if the molecules moving in and out of the cell are:
 - Small
 - Uncharged
 - Move from an area of high concentration to an area of low concentration
- There are three types: diffusion, osmosis, and facilitated diffusion

Diffusion

- Movement of particles from an area of **higher concentration** to an area of lower concentration until an equilibrium is established
- doesn't require energy

Osmosis

- Diffusion of **water across a selectively permeable membrane.**

Facilitated Diffusion

- Movement of **specific molecules** across cell membranes **through protein channels**.
- These specific molecules bind to a protein and are **channeled from the outside of the cell to the inside**
- Is fast and specific
- Will still only occur if **concentration is higher on one side of the membrane than the other**.

Active Transport

- **1. carrier proteins transport substance against its concentration gradient (needs ATP)**
- **a. primary active transport** - energy from ATP used **directly** to transport a substance (e.g. Na^+/K^+ pump, in all cells)
- **b. secondary active transport** - driven by **gradients set up** by primary active transport
- In the digestive tract glucose and amino acids are "dragged along" with Na^+ diffusing into cell (Na^+ gradient set up by Na^+/K^+ pump)

2. vesicular transport (bulk transport)

large molecules or multimolecular substances **enclosed in pieces of membrane**

2.1. Endocytosis

- Process of taking material into the cell by means of **forming a vesicle or pockets of the cell membrane**
- This takes material into the cell and it forms a **vacuole in the cell**

Two Types of Endocytosis

Phagocytosis

- Means “cell eating”
- An extension of cytoplasm surrounds a particle and **package it into a food vacuole**
- requires an enormous amount **of energy**

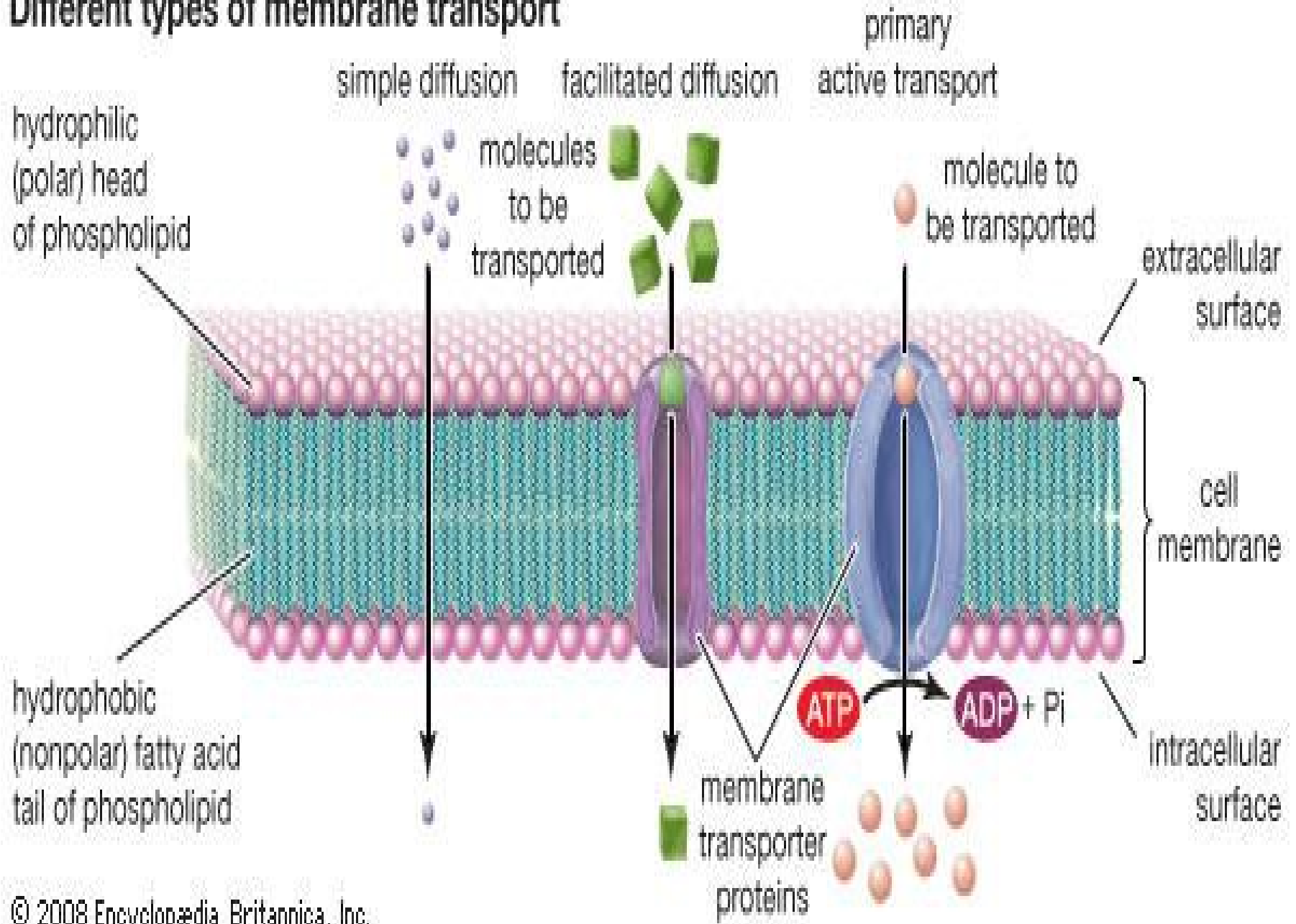
Pinocytosis

- Means “**cell drinking**”
- Tiny pockets form along the cell membrane **fill with liquid** and form vacuoles with in the cell
- Requires enormous amounts of energy

2.2. Exocytosis

- The process where cells **release large amount of materials**
- Vacuole surrounding material fuses with the cell membrane, forcing the contents out of the cell
- Also requires energy

Different types of membrane transport



Intercellular communication

- cells must communicate so they can **coordinate their activities (maintain homeostasis, control growth and development)**
- 3 types of intercellular communication
- **1. gap junctions**
- a. **small molecules and ions** directly exchanged between cells.
- b. important in spread of **electrical signals** (cardiac and smooth muscle, very rarely neurons)

2. signal molecules on cell surface

- . phagocytes (body defense cells)

3. chemical messengers - acts on target cells

(1) paracrines - **act locally** (e.g., histamine in inflammatory response)

(2) neurotransmitters - **act locally**; nerve cells release them to other nerve cells, muscles, or glands

(3) hormones - **acts over long distances**, released into **blood by endocrine glands**

(4) neurohormones - act **over long distances**, released into **blood by special nerve cells** (neurosecretory neuron)

Membrane Potential

1. separated charges have the potential to do work - electrical force of attraction can be harnessed
2. measured in millivolts (mV) - more charges separated → greater potential
3. all plasma membranes have potential
4. due to unequal distribution of a few key ions
 - ions involved
 1. Na^+ & K^+
 2. Cl^-
 3. A^- (Negatively charged anions)
 4. H^+ (proton gradient) – specialized use

Na⁺-K⁺ pump

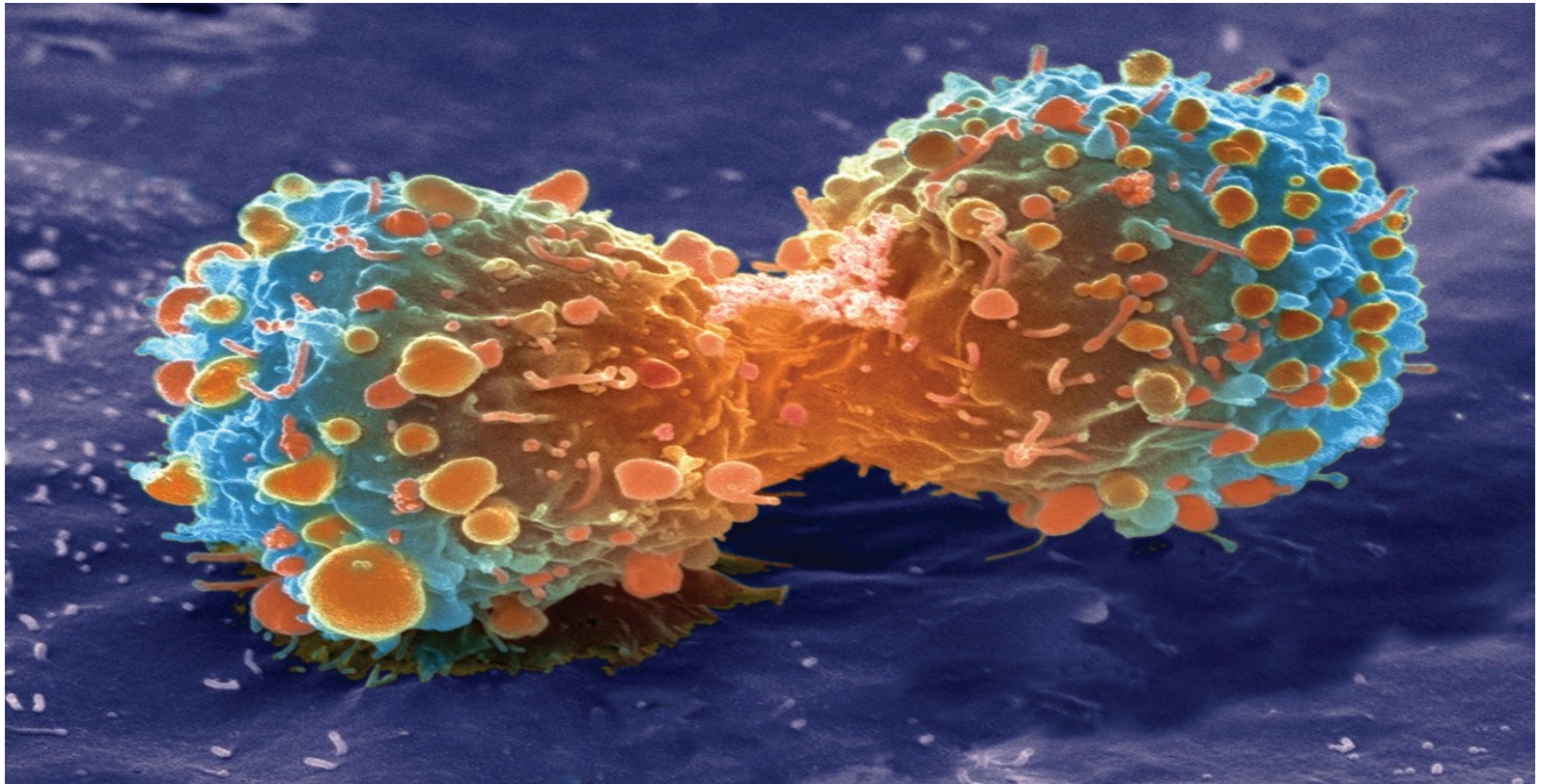
1. responsible both directly and indirectly for **establishing membrane potential**
2. directly generates about 20% of potential
 - A. Actively transports 3 Na⁺ out for every 2 K⁺ in
 - (1) leaves cell slightly negative inside
 - (2) establishes concentration gradients (Na⁺ high outside, K⁺ high inside)
 - B. Passively, Na⁺ → in, K⁺ → out

3. indirectly generates other 80% of membrane potential
 - a. membrane is more permeable to K^+ than to Na^+ (more K^+ channels open)
 - b. K^+ will passively flow out, increasing membrane potential
 - (1) K^+ flows out until concentration gradient is balance by electrical gradient (negative charges inside attract K^+)
 - c. very little Na^+ leaks back in (closed channels)
4. leads to resting membrane potential of $-70mV$ in a typical nerve cell (sign means more negative inside)

The Cell Cycle

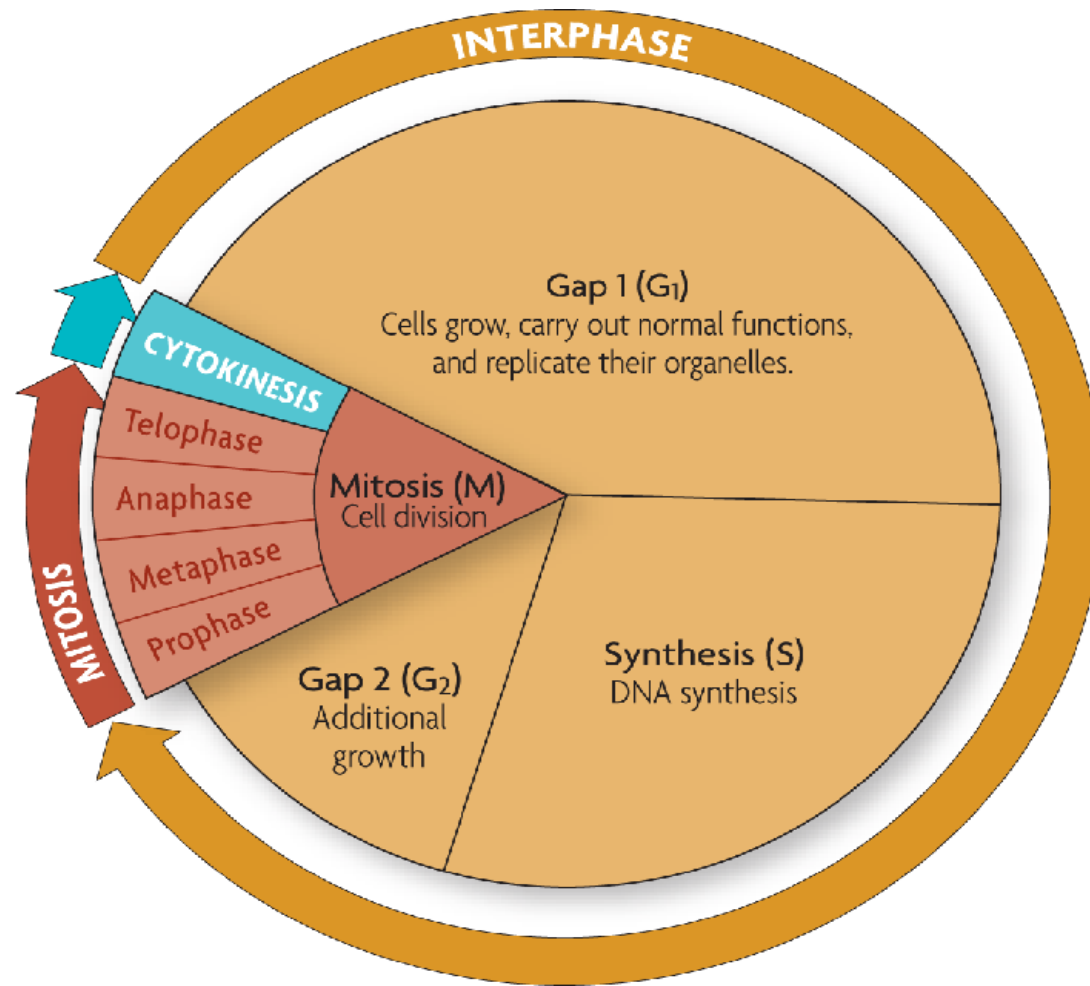
KEY CONCEPT

Cells have distinct phases of growth, reproduction, and normal functions.



•The cell cycle has four main stages

- The cell cycle is a regular pattern of growth, DNA replication, and cell division.



- The main stages of the cell cycle are **gap 1, synthesis, gap 2, and mitosis.**
 - Gap 1 (G_1): cell growth and normal functions
 - DNA synthesis (S): copies DNA
 - Gap 2 (G_2): additional growth
 - Mitosis (M): includes division of the cell nucleus (mitosis) and division of the cell cytoplasm (cytokinesis)
- **Mitosis occurs only if the cell is large enough and the DNA undamaged.**

Cells divide at different rates

FIGURE 5.2 CELL DIVISION

CELL TYPE	APPROXIMATE LIFE SPAN
Skin cell	2 weeks
Red blood cell	4 months
Liver cell	300–500 days
Intestine—internal lining	4–5 days
Intestine—muscle and other tissues	16 years

Regulation of the Cell Cycle

Internal and external factors regulate cell division

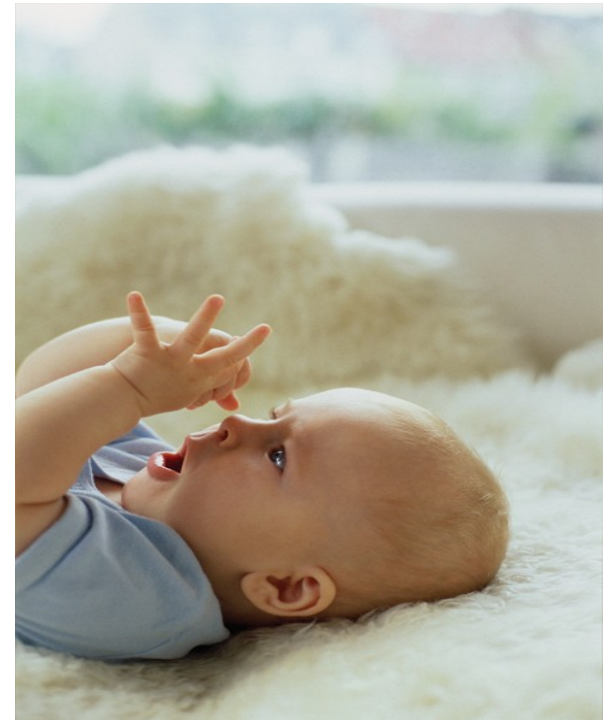
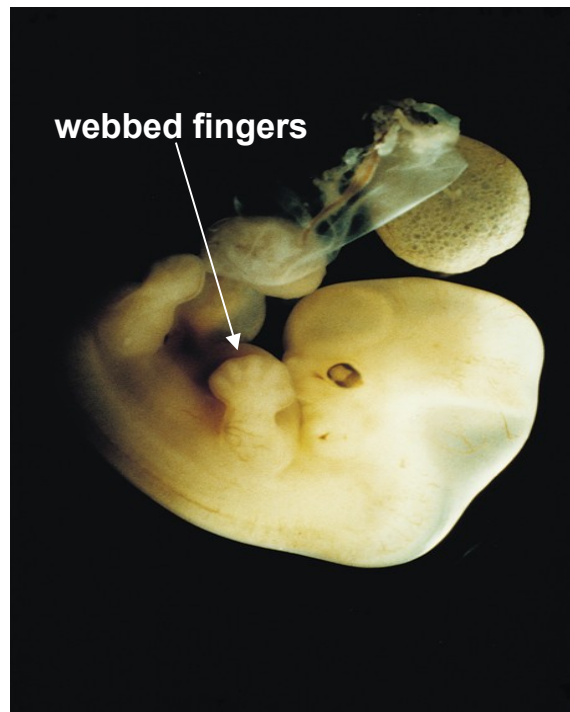
- External factors include physical and chemical signals.
- Growth factors are proteins that stimulate cell division.

Two of the most important internal factors are **kinases and cyclins**.



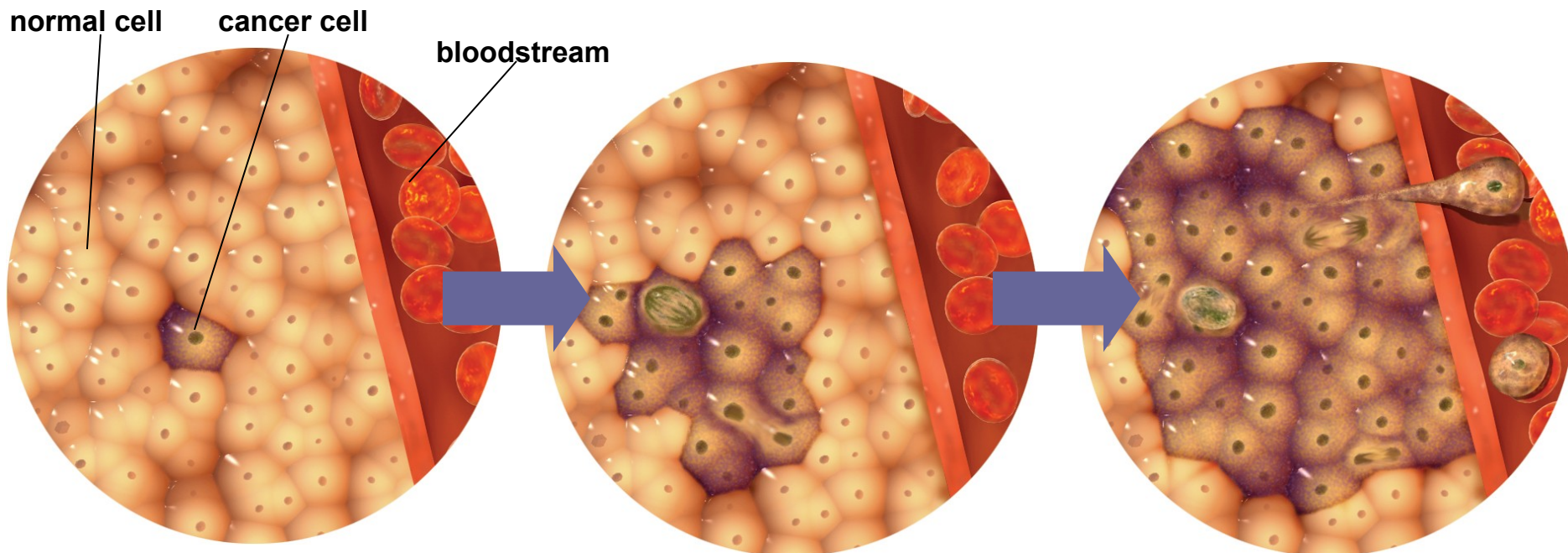
Regulation of the Cell Cycle

- **Apoptosis** is programmed cell death.
 - a normal feature of healthy organisms
 - caused by a cell's production of **self-destructive enzymes**
 - occurs in **development of infants**



Regulation of the Cell Cycle

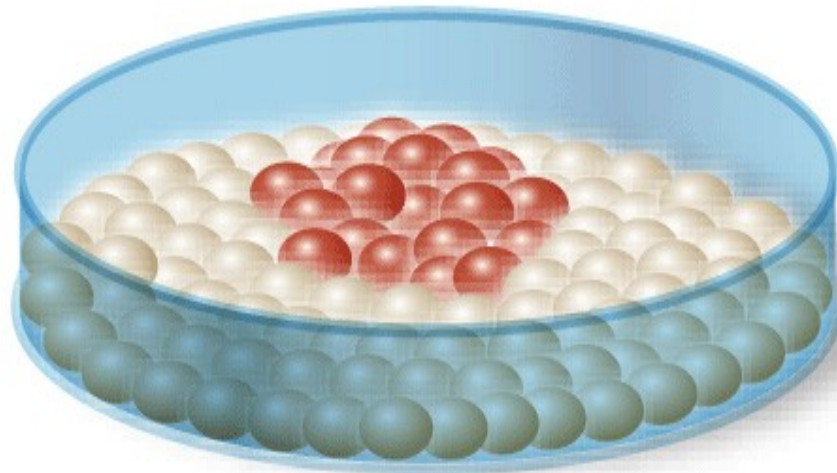
- Cell division is **uncontrolled in cancer**.
- Cancer cells form disorganized clumps called tumors.
 - Benign tumors remain **clustered** and can be removed.
 - Malignant tumors **metastasize**, or break away, and can form more tumors.



Regulation of the Cell Cycle

- Cancer cells do not carry out necessary functions.
- Cancer cells come from normal cells with **damage to genes involved in cell-cycle regulation**.

Cancerous cell growth



Regulation of the Cell Cycle

- **Carcinogens** are substances known to promote cancer.
- Standard cancer treatments typically kill both cancerous and healthy cells.

